

Allen Y. Feng

(423) 767-0200

ayfeng2006@gmail.com

Education:

Duke University

GPA: 3.93

Expected Graduation: May 2028

Skills:

- Languages: Java, C/C++, Python, SQL, HTML, CSS, Fortran 95
- Software: Visual Studio, Spring Boot, Thymeleaf, Git/GitHub, Linux, Docker, Jupyter Notebook, DaVinci Resolve, GIMP
- Courses: Data Structures and Algorithms, Discrete Math, Linear Algebra, Multivariable Calculus, Differential Equations

Experience:

Duke University - Teaching Assistant, Discrete Math (January 2026 - Present):

- Hosted weekly office hours to support students with proof-writing and problem-solving skills in formal logic, set theory, graph theory, and probability.
- Graded assignments and exams, providing clear and constructive feedback.

North Carolina Math Camp - Camp Counselor (July 2025):

- Mentored international high school students in linear algebra, topology, and special relativity, leading group study sessions to develop students' problem solving skills.
- Oversaw and supported campers in a residential setting, maintaining a positive living environment.

Projects:

AI ASL Detector (National AI Summer Camp, 2023):

- Collaborated with a team to train machine learning models for American Sign Language detection by labeling and organizing image datasets of ASL letters, ensuring data quality for model training.
- Produced and edited a promotional video to improve outreach and awareness of the final product.

Newton's Method Calculator:

- Developed a Newton's Method calculator to find roots of single-variable functions, facilitating visualization of iterative mathematical processes.
- Used Spring Boot and Thymeleaf to integrate the calculator into a dynamic webpage, allowing for greater accessibility.
- Implemented algorithms to parse and validate user input of expressions and to evaluate functions at user-defined points.

Activities:

Governor's School for Computational Physics (2022):

- Developed software to model mechanics, thermodynamics, electromagnetism in Fortran 95 using iterative approximation methods, including Euler's method and Fourth-Order Runge-Kutta.
- Learned calculus, differential equations, Linux OS, Fortran 95, machine learning concepts.